

# HKCEE 1993 Mathematics II

93 1. If  $f(x) = 10^{2x}$ , then  $f(4y) =$

- A.  $10^{4y}$ .
- B.  $10^{2+4y}$ .
- C.  $10^{8y}$ .
- D.  $40^y$ .
- E.  $40^{2y}$ .

93 2. If  $s = \frac{n}{2}[2a + (n-1)d]$ , then  $d =$

- A.  $\frac{2(s-an)}{n(n-1)}$ .
- B.  $\frac{2(s-an)}{(n-1)}$ .
- C.  $\frac{s}{n(n-1)}$ .
- D.  $\frac{as-n}{a(n-1)}$ .
- E.  $\frac{4(s-an)}{n(n-1)}$ .

93 3. Simplify  $(x^2 - \sqrt{3}x + 1)(x^2 + \sqrt{3}x + 1)$ .

- A.  $x^4 + 1$
- B.  $x^4 - x^2 + 1$
- C.  $x^4 + x^2 + 1$
- D.  $x^4 - 3x^2 - 2\sqrt{3}x - 1$
- E.  $x^4 + 3x^3 - 2\sqrt{3}x^2 + \sqrt{3}x - 1$

93 4.  $\frac{\sqrt{a}}{\sqrt{a}-\sqrt{b}} + \frac{\sqrt{a}}{\sqrt{a}+\sqrt{b}}$ .

- A.  $\frac{1}{\sqrt{a}-\sqrt{b}}$
- B.  $\frac{a+2\sqrt{ab}-b}{a-b}$

C.  $\frac{\sqrt{b}+\sqrt{a}}{2\sqrt{a}}$

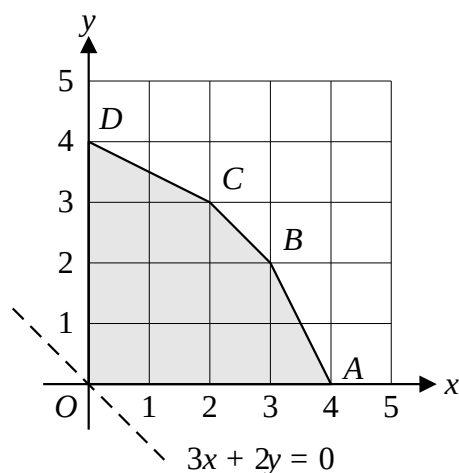
D.  $\frac{b+2\sqrt{ab}-a}{a-b}$

E.  $\frac{a+b}{a-b}$

93 5. If  $3x^2 + ax - 5 \equiv (bx - 1)(2 - x) - 3$ , then

- A.  $a = -5, b = -3$ .
- B.  $a = -5, b = 3$ .
- C.  $a = -3, b = -5$ .
- D.  $a = 5, b = -3$ .
- E.  $a = 3, b = 5$ .

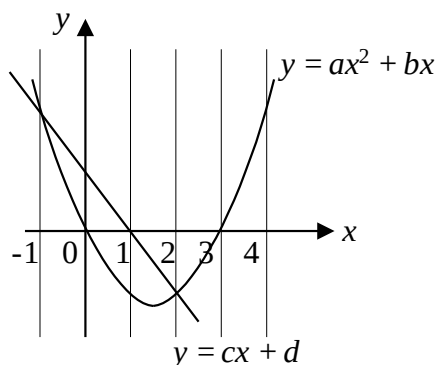
93 6.



Find the greatest value of  $3x + 2y$  if  $(x, y)$  is a point lying in the region  $OABCD$  (including the boundary).

- A. 15
- B. 13
- C. 12
- D. 9
- E. 8

93  
7.



The diagram shows the graphs of  $y = ax^2 + bx$  and  $y = cx + d$ . The solutions of the equation  $ax^2 + bx = cx + d$  are

- A.  $-1, 1$
- B.  $-1, 2$
- C.  $0, 1$
- D.  $0, 3$
- E.  $1, 3$

93  
8. If  $\log(p + q) = \log p + \log q$ , then

- A.  $p = q = 1$ .
- B.  $p = \frac{q}{q-1}$ .
- C.  $p = \frac{q}{q+1}$ .
- D.  $p = \frac{q+1}{q}$ .
- E.  $p = \frac{q-1}{q}$ .

93  
9. The expression  $x^2 - 2x + k$  is divisible by  $(x + 1)$ . Find the remainder when it is divided by  $(x + 3)$ .

- A. 1
- B. 4
- C. 12
- D. 16
- E. 18

93  
10. If 3,  $a$ ,  $b$ ,  $c$ , 23 are in A.P., then  $a + b + c =$

- A. 13.
- B. 26.
- C. 33.
- D. 39.
- E. 65.

93  
11. Find the H.C.F. and L.C.M. of  $ab^2c$  and  $abc^3$

- |    | H.C.F.      | L.C.M.      |
|----|-------------|-------------|
| A. | $a$         | $a^2b^3c^4$ |
| B. | $abc$       | $ab^2c^3$   |
| C. | $abc$       | $a^2b^3c^4$ |
| D. | $ab^2c^3$   | $abc$       |
| E. | $a^2b^3c^4$ | $abc$       |

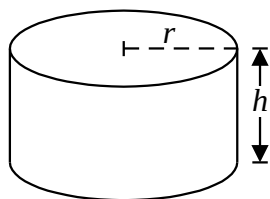
93  
12. If  $\alpha$  and  $\beta$  are the roots of the quadratic equation  $x^2 - 3x - 1 = 0$ , find the value of  $\frac{1}{\alpha} + \frac{1}{\beta}$ .

- A.  $-3$
- B.  $-1$
- C.  $-\frac{1}{3}$
- D.  $\frac{2}{3}$
- E. 3

93  
13. If the simultaneous equations  $\begin{cases} y = x^2 - k \\ y = x \end{cases}$  have only one solution, find  $k$ .

- A.  $-1$
- B.  $-\frac{1}{4}$
- C.  $-4$
- D.  $\frac{1}{4}$
- E. 1

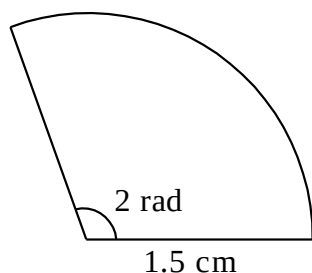
93  
14.



The price of a cylindrical cake of radius  $r$  and height  $h$  varies directly as the volume. If  $r = 5$  cm and  $h = 4$  cm, the price is \$30. Find the price when  $r = 4$  cm and  $h = 6$  cm.

- A. \$25
- B. \$28.80
- C. \$31.50
- D. \$36
- E. \$54

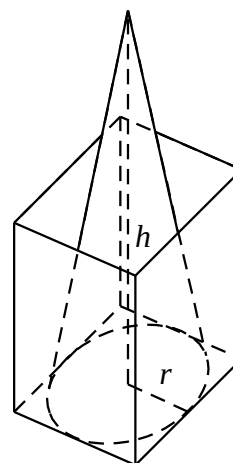
93  
15.



Find the perimeter of the sector in the figure.

- A. 2.25 cm
- B. 3 cm
- C.  $\left(\frac{\pi}{60} + 3\right) \text{ cm}$
- D. 4.5 cm
- E. 6 cm

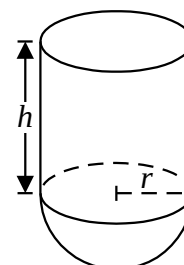
93  
16.



In the figure, the base of the conical vessel is inscribed in the bottom of the cubical box. If the box and the conical vessel have the same capacity, find  $h : r$ .

- A.  $24 : \pi$
- B.  $3 : 1$
- C.  $6 : \pi$
- D.  $3 : \pi$
- E.  $8 : 3\pi$

93  
17.



The figure shows a solid consisting of a cylinder of height  $h$  and a hemisphere of radius  $r$ . The area of the curved surface of the cylinder is twice that of the hemisphere. Find the ratio volume of cylinder : volume of hemisphere

- A.  $1 : 3$
- B.  $2 : 3$
- C.  $3 : 4$
- D.  $3 : 2$
- E.  $3 : 1$

- 93 A merchant marks his goods 25%  
18. above the cost. He allows 10 % discount on the marked price for a cash sale. Find the percentage profit the merchant makes for a cash sale.

A. 12.5%  
B. 15%  
C. 22.5%  
D. 35%  
E. 37.5%

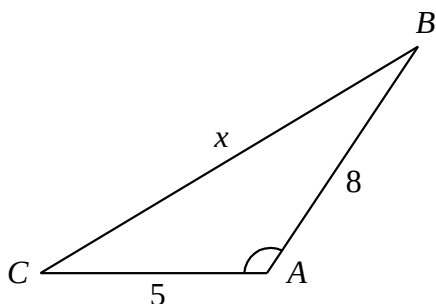
93  
19.  $\frac{\cos \theta}{1 - \sin^2 \theta} \cdot \frac{1 - \cos^2 \theta}{\sin \theta} =$

A.  $\sin \theta$   
B.  $\cos \theta$   
C.  $\tan \theta$   
D.  $\frac{1}{\sin \theta}$   
E.  $\frac{1}{\cos \theta}$

93  
20.  $\cos^4 \theta - \sin^4 \theta + 2 \sin^2 \theta =$

A. 0  
B. 1  
C.  $(1 - \sin^2 \theta)^2$   
D.  $(1 - \cos^2 \theta)^2$   
E.  $(\cos^2 \theta - \sin^2 \theta)^2$

93  
21.



In the figure,  $\cos A = -\frac{4}{5}$ . Find  $a$ .

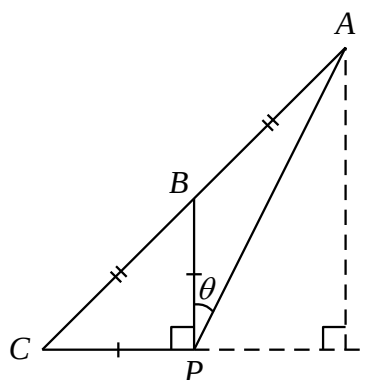
A.  $\sqrt{153}$   
B.  $\sqrt{137}$

C.  $\sqrt{89}$   
D.  $\sqrt{41}$   
E.  $\sqrt{25}$

93 The largest value of  $3\sin^2 \theta + 2\cos^2 \theta -$   
22. 1 is

A. 1.  
B.  $\frac{3}{2}$ .  
C. 2.  
D. 3.  
E. 4.

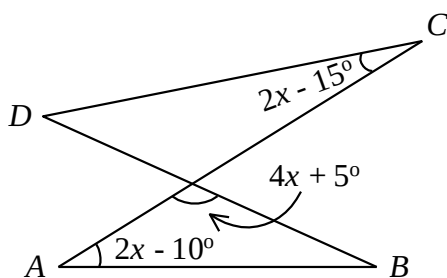
93  
23.



In the figure,  $AB = BC$ ,  $BP = CP$  and  $BP \perp CP$ . Find  $\tan \theta$ .

A.  $\frac{1}{4}$   
B.  $\frac{1}{3}$   
C.  $\frac{1}{2}$   
D.  $\frac{1}{\sqrt{3}}$   
E.  $\frac{\sqrt{3}}{2}$

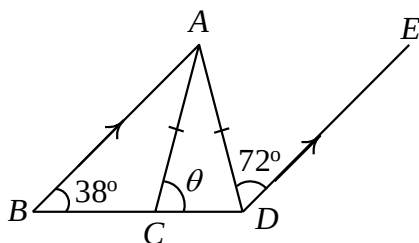
93  
24.



In the figure, points  $A$ ,  $B$ ,  $C$  and  $D$  are concyclic. Find  $x$ .

- A.  $20^\circ$
- B.  $22.5^\circ$
- C.  $25^\circ$
- D.  $27.5^\circ$
- E.  $30^\circ$

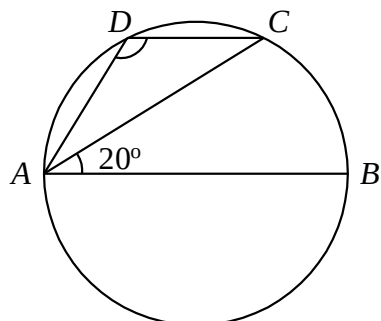
93  
25.



In the figure,  $BA \parallel DE$  and  $AC = AD$ . Find  $\theta$ .

- A.  $34^\circ$
- B.  $54^\circ$
- C.  $70^\circ$
- D.  $72^\circ$
- E.  $76^\circ$

93  
26.



In the figure,  $AB$  is a diameter. Find  $\angle ADC$ .

- A.  $100^\circ$

- B.  $110^\circ$
- C.  $120^\circ$
- D.  $135^\circ$
- E.  $140^\circ$

93 27. If the points  $(1, 1)$ ,  $(3, 2)$  and  $(7, k)$  are on the same straight line, then  $k =$

- A. 3.
- B. 4.
- C. 6.
- D. 7.
- E. 10.

93 28.  $A(0, 0)$ ,  $B(5, 0)$  and  $C(2, 6)$  are the vertices of a triangle.  $P(9, 5)$ ,  $Q(6, 6)$  and  $R(2, -9)$  are three points. Which of the following triangles has/have area(s) greater than the area of  $\triangle ABC$ ?

- I.  $\triangle ABP$
- II.  $\triangle ABQ$
- III.  $\triangle ABR$

- A. I only
- B. II only
- C. III only
- D. I and II only
- E. II and III only

93 29. A circle of radius 1 touches both the positive  $x$ -axis and the positive  $y$ -axis. Which of the following is/are true?

- I. Its centre is in the first quadrant.
- II. Its centre lies on the line  $x - y = 0$ .
- III. Its centre lies on the line  $x + y = 1$ .

- A. I only
- B. II only
- C. III only
- D. I and II only
- E. I and III only

93 30. What is the area of the circle  $x^2 + y^2 - 10x + 6y - 2 = 0$ ?

- A.  $32\pi$

- B.  $34\pi$
- C.  $36\pi$
- D.  $134\pi$
- E.  $138\pi$

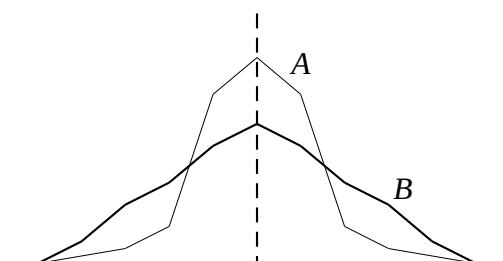
93 Two fair dice are thrown. What is the probability of getting a total of 5 or 10?

- A.  $\frac{1}{9}$
- B.  $\frac{5}{36}$
- C.  $\frac{1}{6}$
- D.  $\frac{7}{36}$
- E.  $\frac{2}{9}$

93 A group of  $n$  numbers has mean  $m$ . If the numbers 1, 2 and 6 are removed from the group, the mean of the remaining  $n - 3$  numbers remains unchanged. Find  $m$ .

- A. 1
- B. 2
- C. 3
- D. 6
- E.  $n - 3$

93  
33.



The figure shows the frequency polygons of two symmetric distributions A and B with the same mean. Which of the following is/are true?

- I. Interquartile range of A < Interquartile range of B
- II. Standard deviation of A >

Standard deviation of B  
III. Mode of A > Mode of B

- A. I only
- B. II only
- C. III only
- D. I and II only
- E. II and III only

93 If  $9^{x+2} = 36$ , then  $3^x =$   
34.

- A.  $\frac{2}{3}$
- B.  $\frac{4}{3}$
- C. 2
- D.  $\sqrt{6}$
- E. 9

93 If  $a : b = 2 : 3$  and  $b : c = 5 : 3$ , then  
35.  $\frac{a+b+c}{a-b+c} =$

- A. -2
- B.  $\frac{5}{2}$
- C. 4
- D.  $\frac{17}{2}$
- E. 31

93  
36.

| $x$   | Sign of $f(x)$ |
|-------|----------------|
| 3.56  | +              |
| 3.58  | -              |
| 3.57  | +              |
| 3.575 | +              |

From the table, a root of the equation  $f(x) = 0$  is

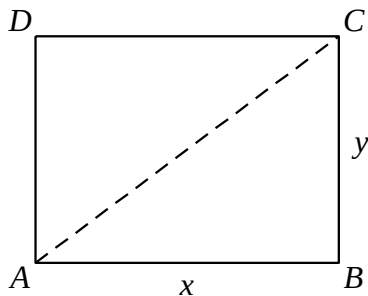
- A. 3.57 (correct to 3 sig. fig.).
- B. 3.575 (correct to 4 sig. fig.).
- C. 3.5775 (correct to 5 sig. fig.).
- D. 3.5725 (correct to 4 sig. fig.).
- E. 3.58 (correct to 3 sig. fig.).

- 93 Given that the positive numbers  $p, q, r,$   
37.  $s$  are in G.P., which of the following  
must be true?

- I.  $kp, kq, kr, ks$  are in G.P., where  $k$   
is a non-zero constant.  
II.  $a^p, a^q, a^r, a^s$  are in G.P., where  $a$   
is a positive constant.  
III.  $\log p, \log q, \log r, \log s$  are in  
A.P.

- A. I only  
B. II only  
C. I and II only  
D. I and III only  
E. I, II and III only

93  
38.



In the figure, the rectangle has  
perimeter 16 cm and area  $15 \text{ cm}^2$ . Find  
the length of its diagonal AC.

- A.  $\sqrt{32}$  cm  
B.  $\sqrt{34}$  cm  
C. 7 cm  
D.  $\sqrt{226}$  cm  
E.  $\sqrt{241}$  cm

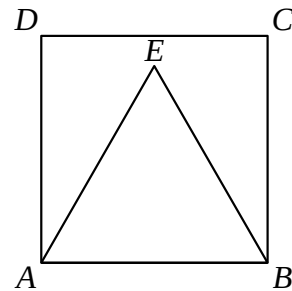
- 93 In factorizing the expression  
39.  $a^4 + a^2b^2 + b^4$ , we find that

- A.  $(a^2 - b^2)$  is a factor.  
B.  $(a^2 + b^2)$  is a factor.  
C.  $(a^2 - ab - b^2)$  is a factor.  
D.  $(a^2 - ab + b^2)$  is a factor.  
E. it cannot be factorized.

- 93 If the solution of the inequality  
40.  $x^2 - ax + 6 \leq 0$  is  $c \leq x \leq 3$ , then

- A.  $a = 5, c = 2$ .  
B.  $a = -5, c = 2$ .  
C.  $a = 5, c = -2$ .  
D.  $a = 1, c = -2$ .  
E.  $a = -1, c = 2$ .

93  
41.

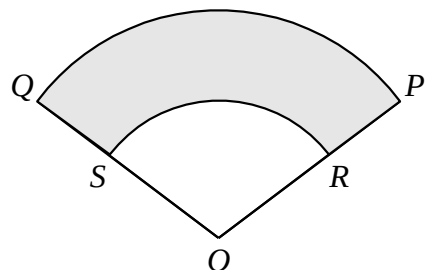


In the figure, ABCD is a square and  
ABE is an equilateral triangle.

$$\frac{\text{Area of } ABE}{\text{Area of } ABCD} =$$

- A.  $\frac{1}{4}$   
B.  $\frac{1}{3}$   
C.  $\frac{\sqrt{3}}{8}$   
D.  $\frac{\sqrt{3}}{4}$   
E.  $\frac{\sqrt{3}}{2}$

93  
42.



In the figure, the radii of the sectors  
OPQ and ORS are 5 cm and 3 cm  
respectively.  $\frac{\text{Area of shaded region}}{\text{Area of sector OPQ}} =$

- A.  $\frac{4}{25}$ .  
 B.  $\frac{2}{5}$ .  
 C.  $\frac{9}{25}$ .  
 D.  $\frac{16}{25}$ .  
 E.  $\frac{21}{25}$ .

93 Which of the following gives the  
 43. compound interest on \$ 10 000 at 6% p.a. for one year, compounded monthly?

- A.  $\$ 10\,000 \times \frac{0.06}{12} \times 12$   
 B.  $\$ 10\,000(1.06^{12} - 1)$   
 C.  $\$ 10\,000 \left(1 + \frac{0.06}{12}\right)^{12}$   
 D.  $\$ 10\,000 \left[\left(1 + \frac{0.06}{12}\right)^{12} - 1\right]$   
 E.  $\$ 10\,000 \left[\left(1 + \frac{0.6}{12}\right)^{12} - 1\right]$

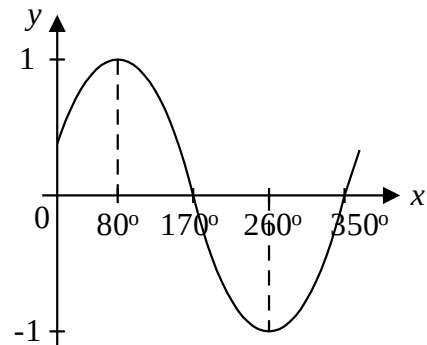
93 Originally  $\frac{2}{3}$  of the students in a class  
 44. failed in an examination. After taking a re-examination, 40% of the failed students passed. Find the total pass percentage of the class.

- A.  $26\frac{2}{3}\%$   
 B.  $33\frac{1}{3}\%$   
 C. 40%  
 D. 60%  
 E.  $73\frac{1}{3}\%$

93 Solve  $\tan^4 \theta + 2\tan^2 \theta - 3 = 0$  for  
 45.  $0^\circ \leq \theta < 360^\circ$ .

- A.  $45^\circ, 135^\circ$  only  
 B.  $45^\circ, 225^\circ$  only  
 C.  $45^\circ, 60^\circ, 225^\circ, 240^\circ$   
 D.  $45^\circ, 120^\circ, 225^\circ, 300^\circ$   
 E.  $45^\circ, 135^\circ, 225^\circ, 315^\circ$

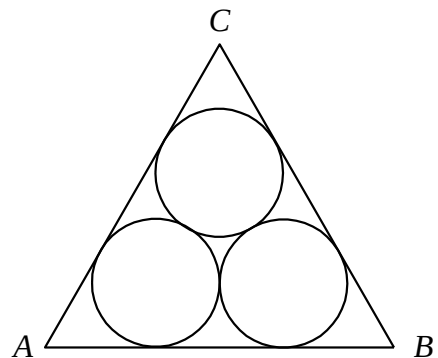
93  
 46.



The figure shows the graph of the function

- A.  $y = \sin(350^\circ - x)$ .  
 B.  $y = \sin(x + 10^\circ)$ .  
 C.  $y = \cos(x + 10^\circ)$ .  
 D.  $y = \sin(x - 10^\circ)$ .  
 E.  $y = \cos(x - 10^\circ)$ .

93  
 47.

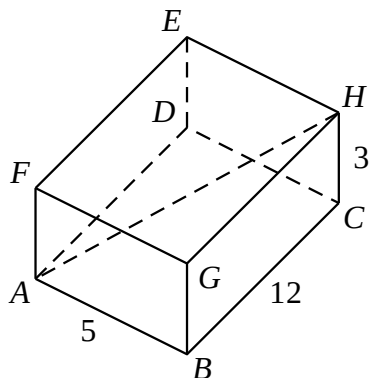


In the figure,  $ABC$  is an equilateral triangle and the radii of the three circles are each equal to 1. Find the perimeter of the triangle.

- A. 12  
 B.  $3(1 + \tan 30^\circ)$   
 C.  $6(1 + \tan 30^\circ)$

- D.  $3\left(1 + \frac{1}{\tan 30^\circ}\right)$   
 E.  $6\left(1 + \frac{1}{\tan 30^\circ}\right)$

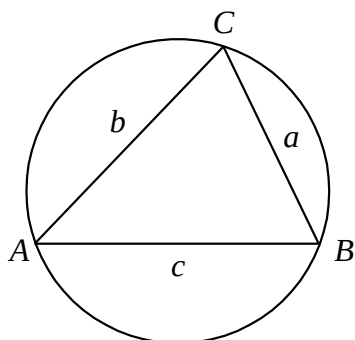
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48.



In the figure,  $ABCDEFGH$  is a cuboid. The diagonal  $AH$  makes an angle  $\theta$  with the base  $ABCD$ . Find  $\tan \theta$ .

- A.  $\frac{3}{5}$   
 B.  $\frac{3}{12}$   
 C.  $\frac{3}{13}$   
 D.  $\frac{3}{\sqrt{178}}$   
 E.  $\frac{\sqrt{153}}{5}$

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49.



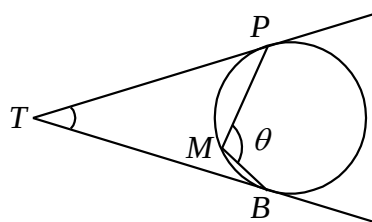
In the figure, if  $\text{arc } BC : \text{arc } CA : \text{arc } AB = 1 : 2 : 3$ , which of the following is/are true?

- I.  $\angle A : \angle B : \angle C = 1 : 2 : 3$

- II.  $a : b : c = 1 : 2 : 3$   
 III.  $\sin A : \sin B : \sin C = 1 : 2 : 3$

- A. I only  
 B. II only  
 C. III only  
 D. I and II only  
 E. I, II and III only

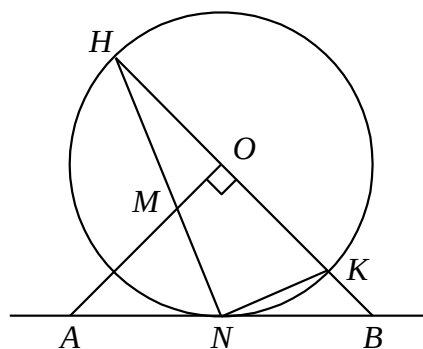
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50.



In the figure,  $TP$  and  $TQ$  are tangent to the circle at  $P$  and  $Q$  respectively. If  $M$  is a point on the minor arc  $PQ$  and  $\angle PMQ = \theta$ , then  $\angle PTQ =$

- A.  $\frac{\theta}{2}$   
 B.  $\theta - 90^\circ$   
 C.  $180^\circ - \theta$   
 D.  $180^\circ - 2\theta$   
 E.  $2\theta - 180^\circ$

93  
51.



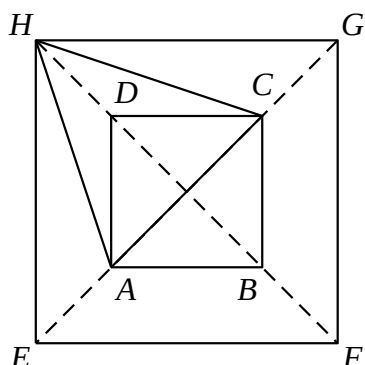
In the figure,  $O$  is the centre of the circle.  $AB$  touches the circle at  $N$ . Which of the following is/are correct?

- I.  $M, N, K, O$  are concyclic.  
 II.  $\triangle HNB \sim \triangle NKB$   
 III.  $\angle OAN = \angle NOB$

- A. I only  
 B. II only

- C. III only  
D. I and II only  
E. I, II and III only

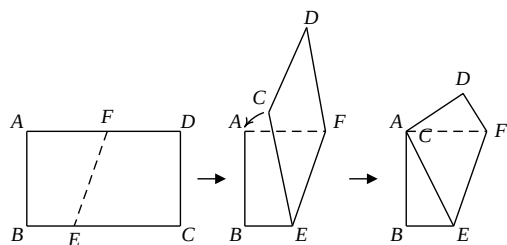
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52.



In the figure  $ABCD$  and  $EFGH$  are two squares and  $ACH$  is an equilateral triangle. Find  $AB : EF$ .

- A.  $1 : 2$   
B.  $1 : 3$   
C.  $1 : \sqrt{2}$   
D.  $1 : \sqrt{3}$   
E.  $\sqrt{2} : \sqrt{3}$

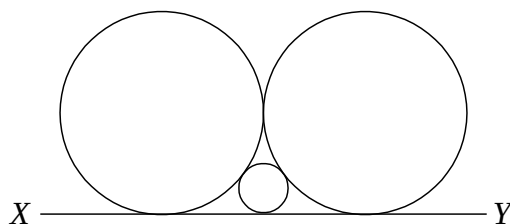
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53.



In the figure, a rectangular piece of paper  $ABCD$  is folded along  $EF$  so that  $C$  and  $A$  coincide. If  $AB = 12$  cm,  $BC = 16$  cm, find  $BE$ .

- A. 3.5 cm  
B. 4.5 cm  
C. 5 cm  
D. 8 cm  
E. 12.5 cm

93  
54.



In the figure, the three circles touch one another.  $XY$  is their common tangent. The two larger circles are equal. If the radius of the smaller circle is 4 cm, find the radii of the larger circles.

- A. 8 cm  
B. 10 cm  
C. 12 cm  
D. 14 cm  
E. 16 cm